SYNTHETIC DOCUMENT PACKET: INS-H105

System Case ID: INS-H105

Domain: Healthcare Medical Review

Authorization Phase: `UNKNOWN` (Clinical Documentation Review)

Document Type: `supporting_medical_report`

Lifecycle Status: Resolved

Sensitivity Classification: Standard (sensitive_case: False)

Assigned Action: `route_to_specialist`

Missing Target Field: None (Semantic Ambiguity / Extraction Defect)

Evidence Reference: `synthetic_history_INS-H105`

Model Confidence Score: `0.42`

Manual Review Required: `True` (Clinical Subject Matter Expert Queue)

Section 1: Referral & Document Overview

- **Case Routing Ref:** SPEC-2026-33910
- **Document Title:** Post-Operative Pathology & Surgical Consultation Narrative
- **Submitting Facility:** Metro Academic Medical Center
- **Patient:** Alan Christopher Blake (DOB: 06/22/1965)
- **Extraction Engine Version:** BioNLP-Parser-v4.1

Section 2: Unstructured Clinical Narrative & OCR Parser Log

Document Narrative Excerpt:

> *"Gross specimen reveals irregular mucosal thickening along the distal segment. Microscopic section: marked mucosal hyperplasia with focal pseudomembranous fibro-adenoid infiltr. w/ atyp. bord., differential diagnosis includes atypical reactive changes vs well-diff neuroendocrine neo. Recommend immunohistochemical staining panel (Chromogranin, Synaptophysin)."*

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| NLP EXTRACTION CONFIDENCE LOG: BIO-NLP-PARSER-V4.1
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| Extracted Entity               SNOMED / ICD-10 Mapping            Model Confidence
Status       |
+-------------------------------------------------------------------------------------------------
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| Mucosal Hyperplasia            36000000 (Hyperplasia of mucosa)    0.91
RESOLVED     |
| pseudomembranous fibro-adenoid [ UNRESOLVED ABBREVIATION / JARGON] 0.38              LOW
CONF     |
| atyp. bord.                    [ AMBIGUOUS PATHOLOGY QUALIFIER ]   0.41              LOW
CONF     |
| Neuroendocrine neo.            371038000 (Neuroendocrine neoplasm) 0.52
BORDERLINE   |
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| OVERALL EXTRACTION CONFIDENCE: 0.42 (< 0.70 Threshold) -> HALT AUTOMATION
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Section 3: Automated Exception & Anomaly Analysis

- **Exception Diagnostic:** `The supporting medical report contained terminology that the standard extraction rules could not interpret reliably.`
- **Defect Classification:** `low_confidence_extraction`
- **Automated Decision Logic:** Extraction confidence score 0.42 is far below the straight-through processing threshold (0.70). Key clinical terms affecting coverage necessity remain ambiguous.
- **Calibrated Uncertainty Guardrail:** The system refuses to hallucinate or probabilistically assign a definitive diagnostic code.

Section 4: Resolution & Reviewer Audit Record

- **Approved Resolution:** Route to a qualified clinical specialist / Medical Review Officer for manual semantic interpretation.
- **Supporting Evidence Rationale:**

> *"Synthetic extraction output left a key term unresolved; the case was routed to a qualified reviewer."*

- **Remediation Trigger:** Case routed to Level-2 Clinical Specialist Queue `QUEUE-MED-SPEC-PATH`: A certified pathology coding specialist reviewed the full narrative and IHC results before entering definitive clinical codes.\n